A unimodal cyclotomic generating function with no prime-index cyclotomic factor

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Task 1 (MathDB 358386). Billey and Swanson's Conjecture 48 in [1] asks whether every nonconstant unimodal basic cyclotomic generating function has a factor $\Phi_p(q)$ for some prime p. A basic cyclotomic generating function is a monic polynomial with constant term 1 and nonnegative integer coefficients whose roots are roots of unity. Their monoid of unimodal such polynomials is denoted by Φ^{uni}. We disprove the conjecture with an explicit degree-66 polynomial. The statement occurs on page 32 of the final article and page 33 of arXiv:2305.07620v3.

Theorem 1. *The polynomial*

$$F(q) = \Phi_4(q)^7 \Phi_9(q)^4 \Phi_{25}(q) \Phi_{30}(q)$$

belongs to Φ^{uni}*, is nonconstant, and has no factor* $\Phi_p(q)$ *with* p *prime.*

Proof. The four cyclotomic polynomials in the displayed product are

$$\Phi_4(q) = 1 + q^2, \qquad \Phi_9(q) = 1 + q^3 + q^6,$$
$$\Phi_{25}(q) = 1 + q^5 + q^{10} + q^{15} + q^{20},$$
$$\Phi_{30}(q) = 1 + q - q^3 - q^4 - q^5 + q^7 + q^8.$$

These identities follow, for example, by exact division in $q^n - 1 = \prod_{d \mid n} \Phi_d(q)$. Each factor is reciprocal, so F is reciprocal of degree $7 \cdot 2 + 4 \cdot 6 + 20 + 8 = 66$. Write $F(q) = \sum_{s=0}^{66} c_s q^s$. The coefficients through the midpoint are as follows; the others are given by $c_{66-s} = c_s$.

s	c_s	s	c_s	s	c_s	s	c_s
0	1	9	198	18	920	27	1450
1	1	10	283	19	940	28	1570
2	7	11	337	20	1025	29	1588
3	10	12	437	21	1058	30	1704
4	24	13	524	22	1109	31	1737
5	42	14	592	23	1184	32	1746
6	63	15	701	24	1220	33	1810
7	105	16	763	25	1319		
8	147	17	829	26	1381		

For completeness, the following finite sum specifies every entry of the table directly. Define

$$b_s = \sum_{i=0}^{7} \sum_{j=0}^{4} \sum_{k=0}^{4-j} \sum_{h=0}^{4} \binom{7}{i} \frac{4!}{j!k!(4-j-k)!} \mathbf{1}_{\{s=2i+3j+6k+5h\}},$$

so b_s is the coefficient of q^s in $(1+q^2)^7(1+q^3+q^6)^4(1+q^5+q^{10}+q^{15}+q^{20})$, with $b_s = 0$ outside $0 \le s \le 58$. Multiplication by Φ_{30} gives

$$c_s = b_s + b_{s-1} - b_{s-3} - b_{s-4} - b_{s-5} + b_{s-7} + b_{s-8}.$$

This proves the tabulated values by finite integer arithmetic. They satisfy $1 = c_0 \le c_1 \le \cdots \le c_{33}$. Reciprocity now proves that every coefficient is positive and that the entire sequence is unimodal. The factorization also proves that F is monic with constant term 1 and that every root is a root of unity. Thus $F \in \Phi^{\mathrm{uni}}$.

Finally, every root of F has exact order in $\{4, 9, 25, 30\}$. If Φ_p divided F for a prime p, a primitive pth root of unity would be a root of F. None of the four possible orders is prime, which is a contradiction. Since $\deg F = 66 > 0$, this is a counterexample to the stated conjecture. □

Verification and scope. The accompanying standard-library Python verifier constructs cyclotomic polynomials from the divisor identity, checks the table by two exact expansions, and checks nondivisibility by Φ_p for every prime $p \le 67$ by polynomial division. Larger primes are excluded by degree. An additional SymPy expansion independently reproduced all 67 coefficients. No claim is made that degree 66 is minimal. This example is not log-concave: $c_1^2 = 1 < 7 = c_0 c_2$. The source's prime-power factor theorem is consistent with the example, which has factors $\Phi_4, \Phi_9, \Phi_{25}$.

Proposed Tasks 2–5. The following directions are proposed after the completed resolution of Task 1; they are not claimed to be solved here, and their novelty has not been audited.

2. Determine the smallest degree of a nonconstant member of Φ^{uni} having no prime-index cyclotomic factor, with a certificate excluding every smaller degree.

3. Characterize the nonnegative integer quadruples (a, b, c, d) for which $\Phi_4^a \Phi_9^b \Phi_{25}^c \Phi_{30}^d$ is unimodal.

4. Give an algorithm that, for an input degree bound D, enumerates exactly the irreducible elements of Φ^{uni} of degree at most D that have no prime-index cyclotomic factor, and prove its correctness.

5. Let $N(D)$ count the nonconstant elements of Φ^{uni} of degree at most D having no prime-index cyclotomic factor. Determine the asymptotic order of $\log N(D)$ as $D \to \infty$.

References

[1] S. C. Billey and J. P. Swanson, *Cyclotomic Generating Functions*, Electronic Journal of Combinatorics **31**(4) (2024), P4.4, 43 pp. doi:10.37236/12687. Current manuscript: arXiv:2305.07620v3.